

# Cross-Modal Recipe Retrieval

Aligning Food Images and Long-Form Recipes via Lightweight Adapters and  
Multi-Modal Alignment

# | Curating the Dataset: Filtration



# | DAR: SAM & Visual Imagination



## Segment Anything (SAM)

Crops central food items to suppress background noise like tables and plates.

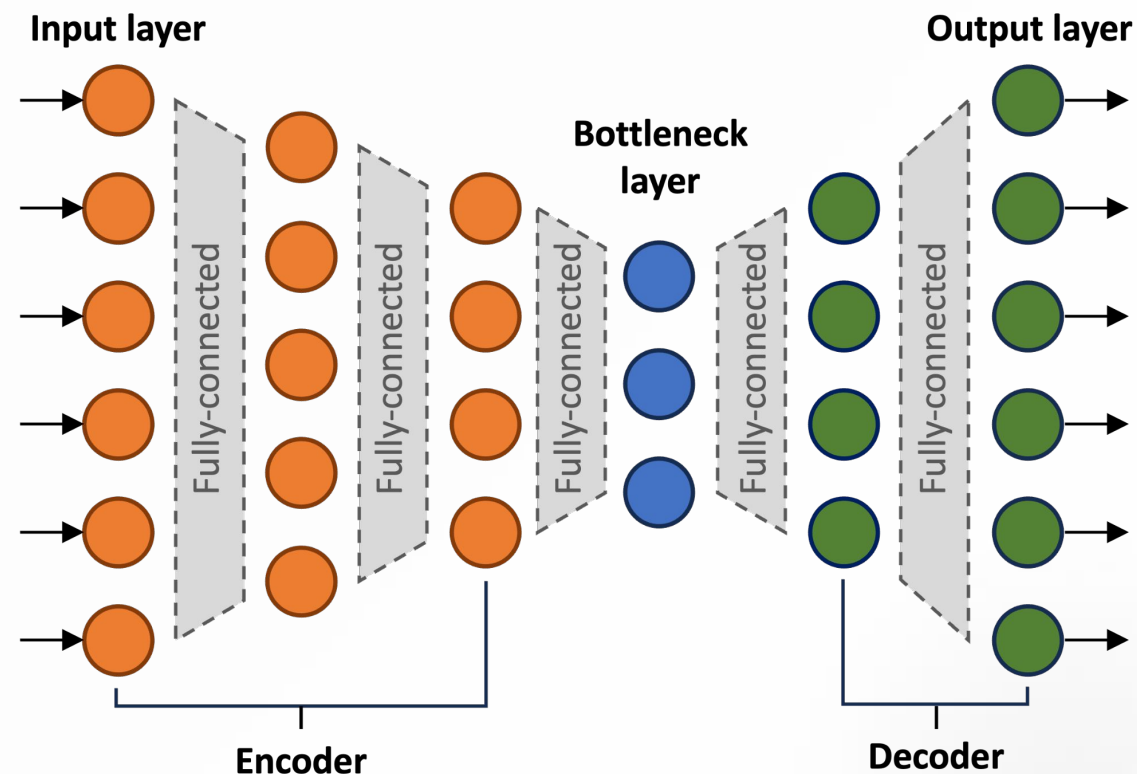
## LLM Visual Imagination

Llama 3 translates ingredients into ~30-word visual descriptions (plating, texture).

# Parameter-Efficient Adapters

## Architectural Philosophy

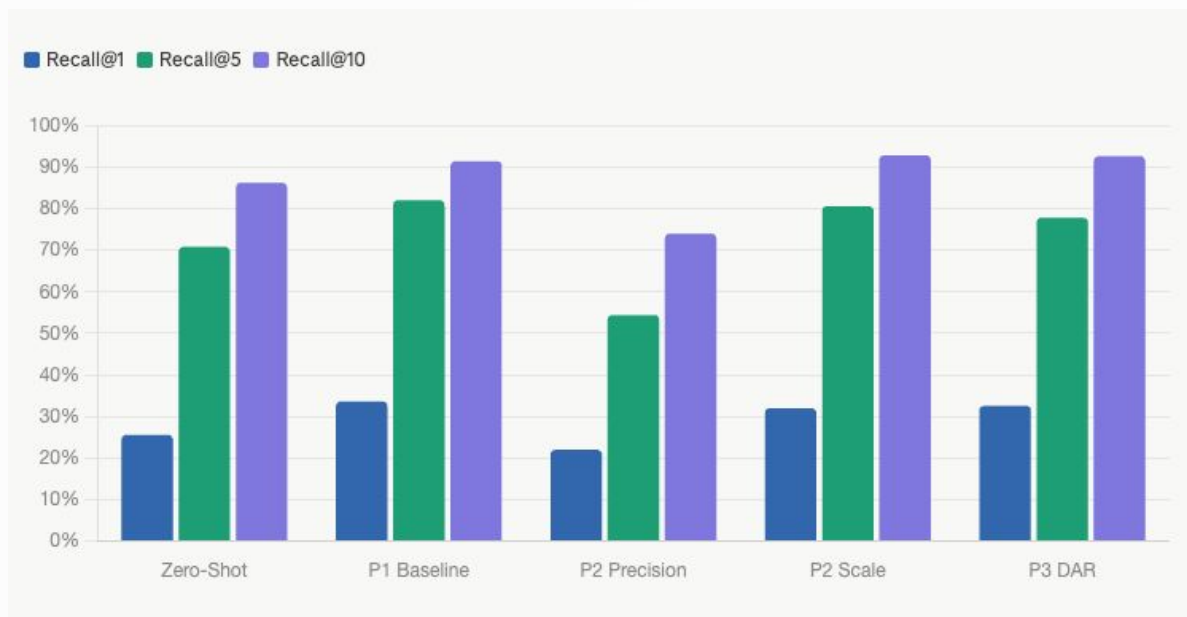
- ✓ **Frozen Encoders:** Base LongCLIP weights are locked to prevent overfitting.
- ✓ **Bottleneck Adapters:** Lightweight MLP layers (64 or 256 dim) mapped to shared space.
- ✓ **Zero Initialization:** Output layer starts at zero for stable identity mapping.
- ✓ **Loss Functions:** Shift from InfoNCE to Multi-Level Circle Loss.



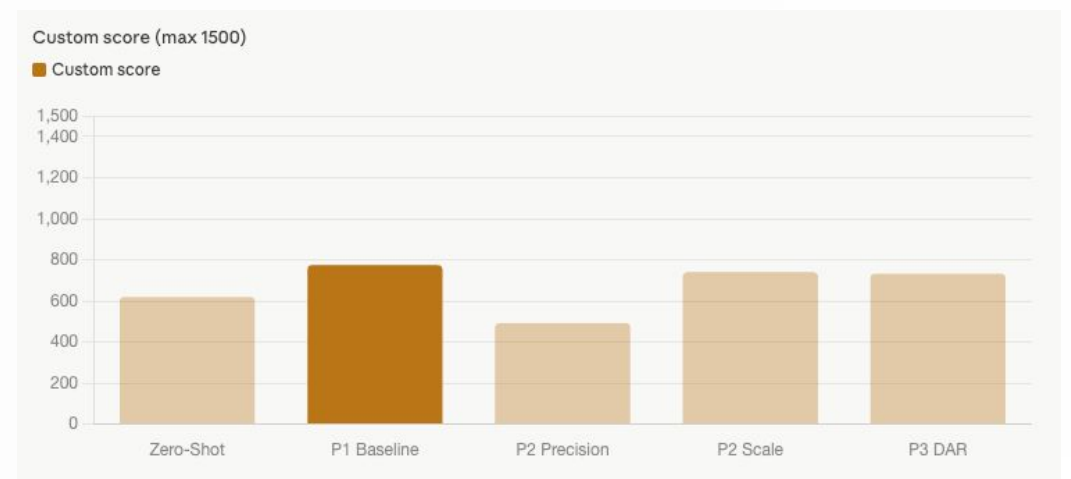
# Adapter Setup & Parameters

| Configuration     | Bottleneck Dim | Dataset Size | Loss Function           |
|-------------------|----------------|--------------|-------------------------|
| Phase 1 Baseline  | 256            | 5,000        | InfoNCE                 |
| Phase 2 Precision | 64             | 8,000        | InfoNCE                 |
| Phase 2 Scale     | 64             | 74,000       | InfoNCE                 |
| Phase 3 DAR       | 64             | 74,000       | Multi-Level Circle Loss |

# Quantitative Performance

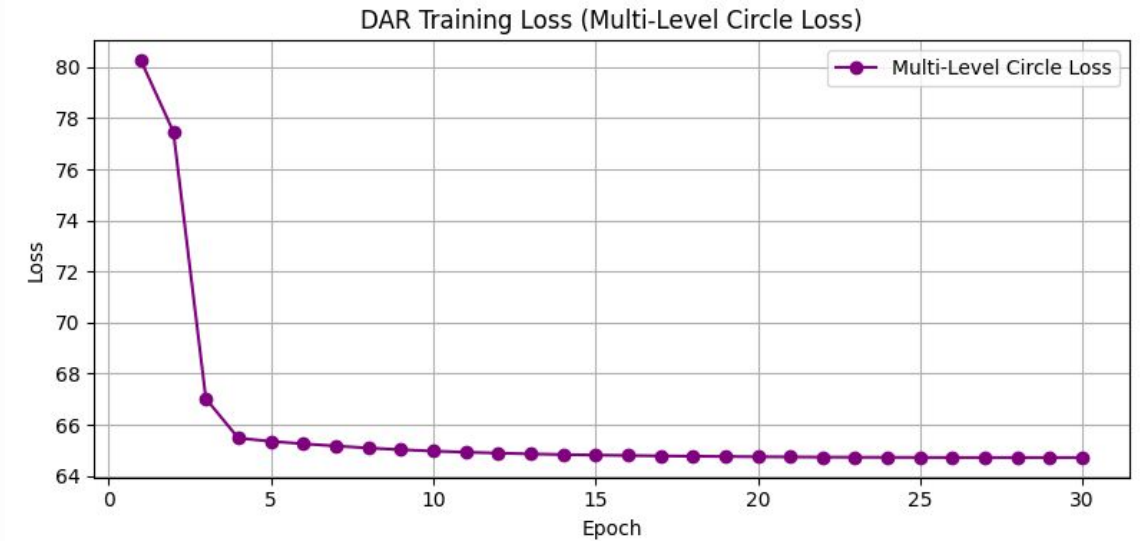
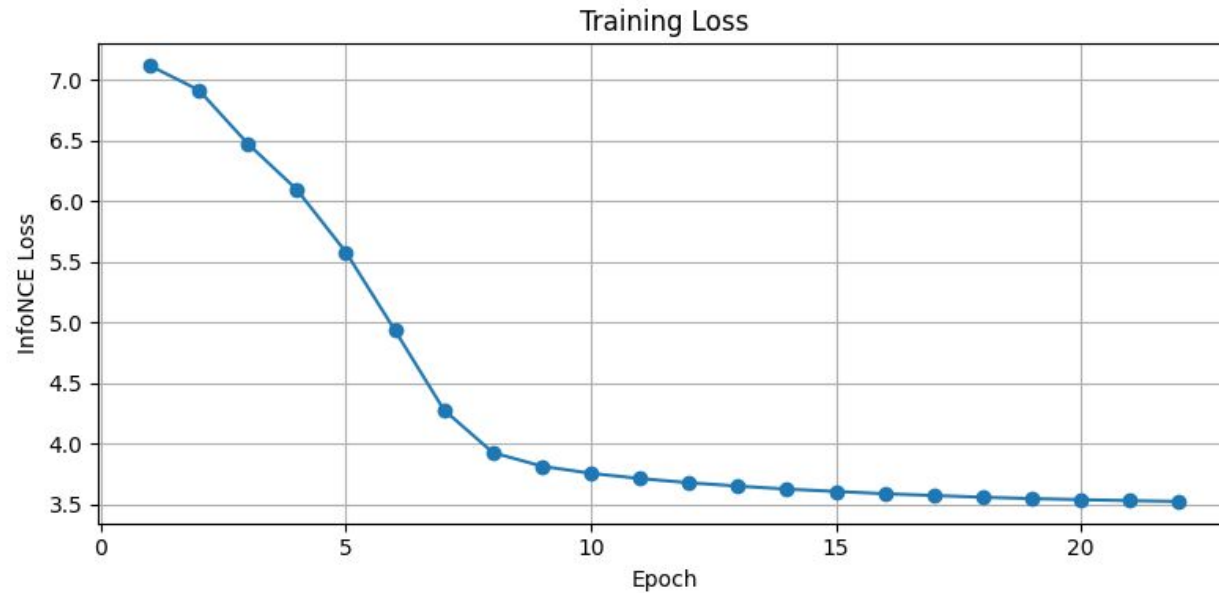


Recall@1, @5, @10 Comparison



Custom Weighted Score

# Saturation Dynamics



**8k Dataset:** ~6 Minutes training duration.

**74k Dataset:** ~1.5 Hours training duration.

**DAR Phase:** ~3.7 Hours (2.5x slower) including augmentation

# Output Space Retrieval

## Pepperoni Pizza Query

Example of successful cross-modal alignment on a 2 million recipe pool.

- ✓ Rank 1: Three-Meat Calzone
- ✓ Rank 2: Pizza Margherita with A Black Truffle ...
- ✓ Rank 3: Margherita Pizza With Tomato ...

Demonstrates semantic proximity between visually similar dishes (calzones vs pizzas).

Query: peperoni-pizza-blogbeitrag.jpg



#1 Three-Meat Calzone (score: 0.357)

6 ounces salami, thinly sliced, 2 ounces pepperoni, thinly sliced, 1 cup ricotta cheese, 1/2 cup parmesan cheese, grated, 1/2 cup fresh basil, chopped  
["Adjust oven rack to upper-middle position and heat oven to 450 degrees. Arrange salami and pepperoni on microwave-safe plate lined with paper towels. Cover wi...

#2 Pizza Margherita With A Black Truffle Oil Crust (score: 0.352)

1/2 Pillsbury Gluten-Free Thin Crust Dough, 4 ounces Tomato Sauce, 4 ounces Fresh mozzarella, sliced, 2 tablespoons Black truffle oil, 2 teaspoons Red pepper flakes  
["Preheat oven to 400 degrees and grease baking sheet.", "Lay down pizza dough (approx half of the container) on baking sheet and roll out until thin.", "Brush ...

#3 Margherita Pizza With Tomato, Mozzarella And Basil (score: 0.346)

One portion Perfect Neapolitan Pizza Dough, 1/2 cup Perfect Pizza Sauce, 1/4 pound thinly sliced fresh mozzarella cheese, 10 fresh basil leaves, Extra-virgin olive oil, for drizzling  
["Set a pizza stone on a rack in the top third of the oven.", "Preheat the oven to 500 for at least 45 minutes.", "Meanwhile, remove the dough ball from the ref...



# Methodological Challenges



## 77-Token Wall

CLIP's native limit is too short for recipes. Solved by migrating to LongCLIP (248 tokens).



## Bottleneck Loss

64-dim adapters limit representation. Solved by identifying 256-dim as the performance sweet spot.



## Granularity of Categories

The model struggled to discriminate visually similar but distinct categories because the training labels lacked visual granularity. Solved by implementing DAR framework

# | Overall Learnings

- ✓ **Capacity Over Scale:** Adapter width is more critical than data volume alone. We recommend a 256+ dimension bottleneck for complex culinary text.
- ✓ **DAR Effectiveness:** SAM background suppression and Multi-Level Circle Loss effectively align fine-grained visual/imagined modalities.
- ✓ **Next Steps:** Unfreeze the last 1–2 layers of the LongCLIP base model to allow direct vocabulary adaptation for better Recall@1.

**Thank you for your attention!**

Questions?